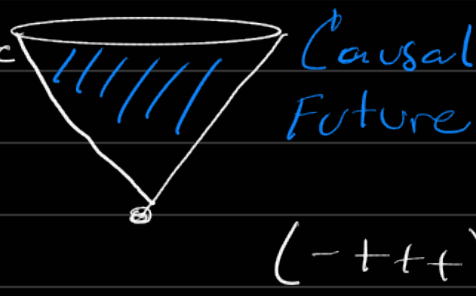


* Given a manifold with Riemann metric (M, g_{ab}) , at each point $p \in M$ the tangent space $T_p M \cong M^n$ is isomorphic to Minkowski $(-+++)$



• Light cone at the origin of $T_p M$
 $\Rightarrow L.C. \subset T_p M$

• Non simply connected spaces may not be time orientable

Defn (M, g) is time orientable if one can make a globally consistent choice of future direction. $\forall p \in M$

$\exists V \in \Gamma(TM)$ s.t. $g(V, V) < 0$ everywhere

• $-t^2 + x^1 + x^2 + x^3^2 < 0 \Rightarrow |t| > |x|$

• $> 0 \Rightarrow |x| > |t|$

* Given any such V s.t. $g(V, V) < 0$, then we find: $u \in T_p M$:

$g(V, u) < 0 \Rightarrow u$: Future directed

Thm: Let (M, g) be time orientable, then $\exists$ a (highly unique) smooth non-vanishing timelike vector field t^a on M .

proof: Since M is paracompact, we can choose a Riemannian metric h_{ab} on M (comes from the existence of a partition of unity: can give local $\mathbb{R}$ data to create a global smooth pos definite metric on all of M).

• At each $p \in M$ there will be a unique future directed timelike vector t^a which minimizes the value of $g_{ab} V^a V^b$ for vectors subject to $h_{ab} V^a V^b = 1$

• Cover M by $(\mathcal{U}_\alpha, \mathcal{O}_\alpha)$ and $\overline{\mathcal{O}_\alpha}$ compact
 $\Rightarrow$ locally $g_{ab} V^a V^b$

* $(\mathcal{O}_\alpha, \mathcal{U}_\alpha)$: with local metric $(g_\alpha)_{ab}$

$(g_\alpha)_{ab} := dx^i_\alpha \otimes dx^i_\alpha, i=1, \dots, n, \quad V = V^i \partial x^i_\alpha$

$(g_\alpha)(V, V) = \sum_i 1 v^i \otimes dv^i(V, V)$

$$\begin{aligned}
 &= \sum_{i=1}^n dx_a^i \left(V^m \frac{\partial}{\partial x_a^m} \right) \cdot dx_a^i \left(V^n \frac{\partial}{\partial x_a^n} \right) \\
 &= \sum_{i=1}^n V^m \delta_m^i V^n \delta_n^i = \sum_{i=1}^n (V^i)^2 \quad \forall V \neq 0 \\
 &\geq 0
 \end{aligned}$$

$\leadsto$ P.O.D. $\{f_\alpha\}$ $f_\alpha: \mathcal{O}_\alpha \rightarrow \mathbb{R}$

- (i) $\text{Supp}\{f_\alpha\} \subset \mathcal{O}_\alpha$
- (ii) $0 \leq f_\alpha \leq 1$
- (iii) $\sum_\alpha f_\alpha = 1$

Define $g_{ab} := \sum_\alpha f_\alpha (g_\alpha)_{ab}$

- Unique future directed section t^a , (K_{ab} on M)
 $K_{ab} t^a t^b = 1$

* Full proof: At $p \in M$ consider the n -unit sphere in $T_p M$: $\|t\|^2 = \langle t, t \rangle = K_{ab} t^a t^b = 1$

$$S_p := \{v^a \in T_p M \mid K_{ab} v^a v^b = 1\}$$

S_p is closed/bounded $\Rightarrow$ compact and since K_{ab} is Euclidean $\Rightarrow K_{ab} \geq 0 \quad \forall t^a/v^a$.

- Map $V \mapsto g_{ab} v^a v^b$ is continuous on S_p and obtains a min/max.

$$T_p M \rightarrow \mathbb{R}: \|\cdot\|,$$

- define K -self adjoint endo:

$$A_b^a := K^{ac} g_{cb} \in T_p M \otimes T_p^* M$$

$$\Rightarrow K_{ac} A_b^c = g_{ab}$$

$$A_b^a v^b = K^{ac} g_{cb} v^b = K^{ca} v_c = \tilde{v}^a$$

$$\Rightarrow g_{ab} v^a v^b = K_{ac} A_b^c v^a v^b = K(Av, v) = g(v, v)$$

A is self adjoint, i.e. $K(Av, v) = K(v, Av)$

$\leadsto$ Expand V in an eigen basis of A as an operator

- $Av = \lambda v$, on $\mathcal{U}^0$, $\lambda_0 < 0$, $\lambda_i \geq 0 \quad \forall i \geq 1$

$$v = c_i e_i, \quad \sum_i c_i^2 = 1$$

$$\begin{aligned}
 g(v, v) &= K(v, Av) = K(v, \lambda v) = \sum_{ij} c_i c_j K(e_i, e_j) \lambda_j \\
 &= \sum_{ij} c_i c_j \lambda_j \delta_{ij} = \sum_i c_i^2 \lambda_i
 \end{aligned}$$

$$\Rightarrow g(v, v) = \sum_i c_i^2 \lambda_i$$

minimized for $\min \{ \lambda_i \}$, i.e. λ_0

or V is in the time like direction

$$V \geq \pm e_0 \quad (\text{Both timelike})$$

$$g(V, V) = g(e_0, e_0) = \lambda_0 < 0$$

$-e_0$: future directed

$+e_0$: Past directed

$$\Rightarrow \boxed{t^a(p) := +e_0}$$

* $\forall p \in M$, $\exists!$ future directed unit time vector $k_{ab} t^a t^b = 1$ which minimizes $g_{ab} v^a v^b$

• Smoothness: Lagrange multiplier eq.

$$\hookrightarrow g_{ab} t^b = \lambda k_{ab} t^b$$

$$\hookrightarrow k_{ac} A^c_b t^b = \lambda k_{ac} t^c$$

• $\lambda_0(p)$ and its eigenspace vary smoothly with p .

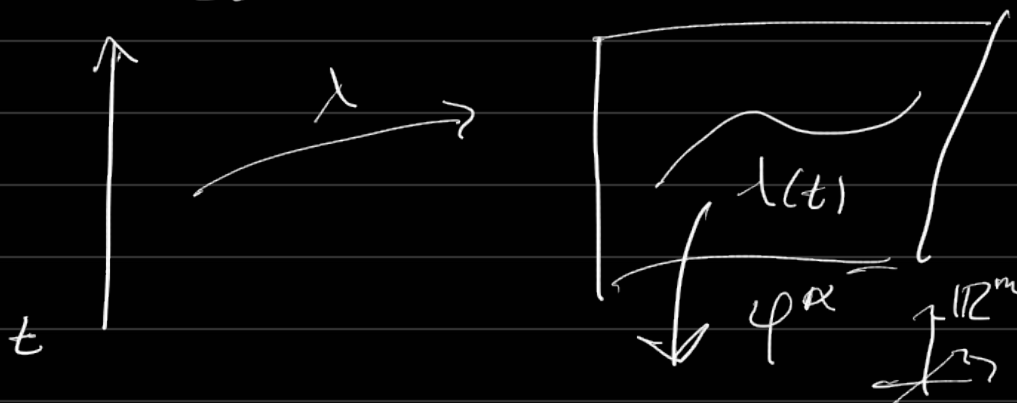
$$\hookrightarrow e_0^a \mapsto e_0^a / \sqrt{k_{ab} e_0^a e_0^b}$$

(if it is not already normalized)

$\Rightarrow$ Lemma: If a continuous, timelike vector field exists then (M, g_{ab}) is time orientable.

* Let (M, g) be time orientable. A differentiable curve $\lambda(t)$ is said to be future directed timelike curve if at each pt $p \in \lambda$ the tangent t^a is future directed timelike vec.

$$V = \frac{d\lambda(t)}{dt}$$



$$x^\mu = (x^1(t), \dots, x^n(t))$$

$$\lambda(t) \in T_{\lambda(t)} M \mid V_{\lambda(t)} = \dot{\lambda}(t)$$

$$\varphi_* \lambda(t) \equiv x^\mu(t) = (x^1(t), \dots, x^n(t))$$

$$\frac{d}{dt} ((\varphi^{-1})_* (\varphi_* \lambda)(t)) = \frac{d}{dt} (\varphi^{-1}_* (x^\mu(t)))$$

$$= \varphi_* \lambda(t) \quad \lambda(t) = \dots$$

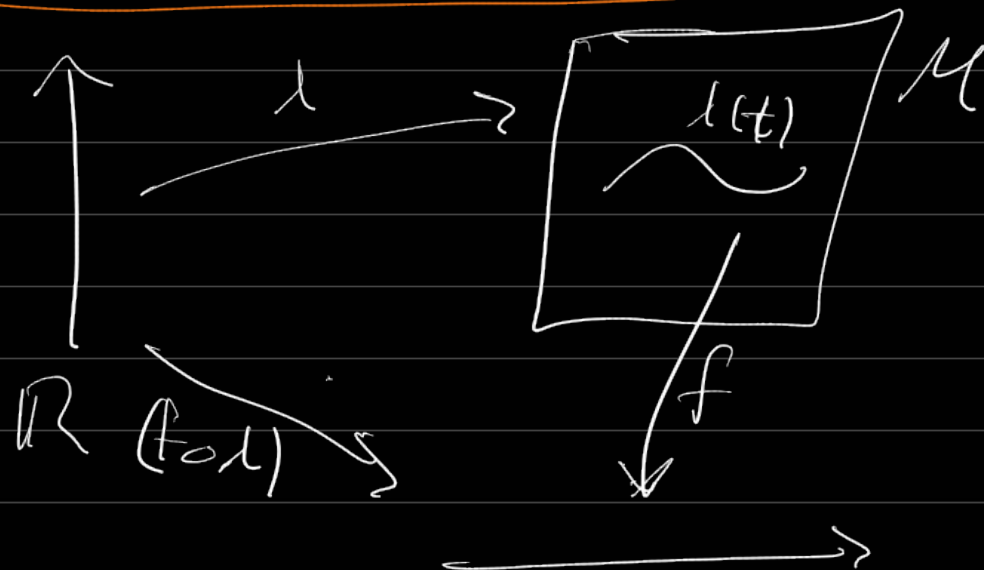
$$= \frac{\partial \lambda}{\partial t} \frac{\partial}{\partial x^0} (\psi(x^n(t)))$$

* Def: $\lambda(t) = (\dot{x}^1(t), \dot{x}^2(t), \dots, \dot{x}^n(t))$
 here the basis is $\left\{ \frac{\partial}{\partial x^1} \lambda(t), \frac{\partial}{\partial x^2} \lambda(t), \dots \right\}$
 so $\frac{\partial}{\partial x^1} = (1, 0, \dots, 0)$ etc

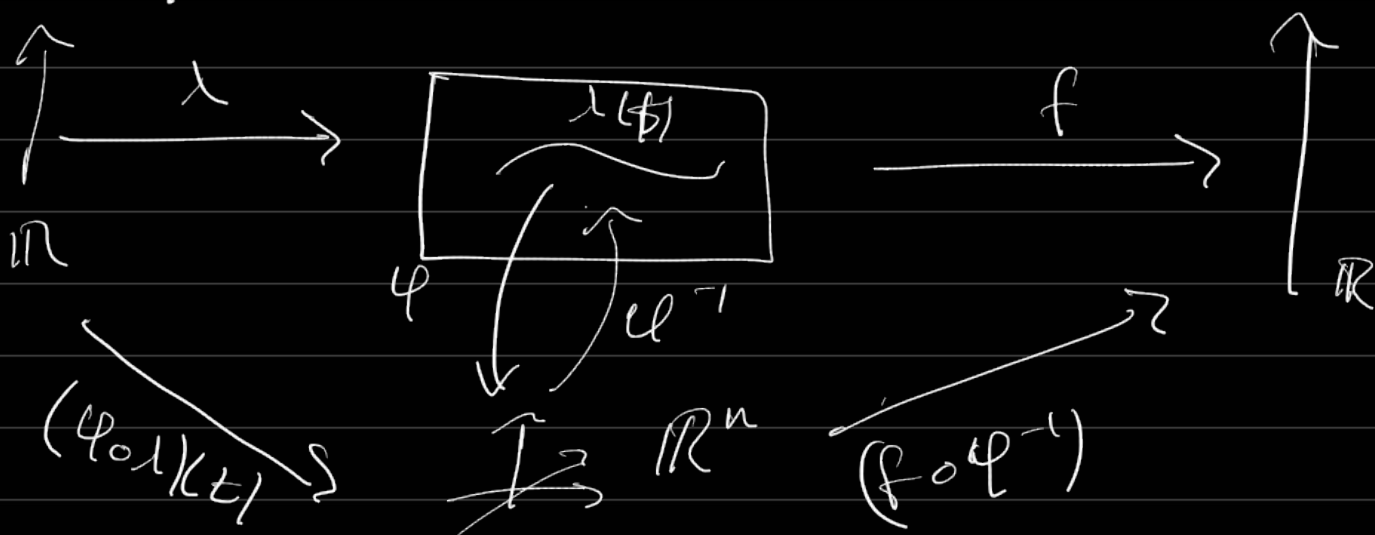
Smooth Curve: $\lambda \in C^\infty(\mathbb{R}) : \mathbb{R} \rightarrow M$

$t \in T_{\lambda(t)} M$. Let $f \in F(M)$, $T \in T_p M$

$$T(t) = \frac{d}{dt} (f \circ \lambda) : \mathbb{R} \rightarrow \mathbb{R}$$



$$T(t) = \frac{d}{dt} (f \circ \lambda)(t) = \frac{d}{dt} (f \circ \psi^{-1}) \circ (\psi \circ \lambda)(t)$$



$$\begin{aligned} (\psi \circ \lambda)(t) &= x^n(t) \\ \frac{d}{dt} (f \circ \psi^{-1})((\psi \circ \lambda)(t)) &= \frac{\partial (f \circ \psi^{-1})}{\partial x^n} \cdot \frac{d(\psi \circ \lambda)^n}{dt} \\ &= \frac{\partial}{\partial x^n} (f \circ \psi^{-1}) \frac{dx^n}{dt} \end{aligned}$$

but $\dot{x}^\mu = \partial_\mu (f \circ \phi^{-1})$

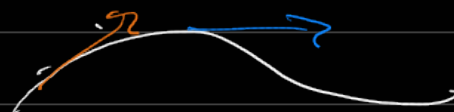
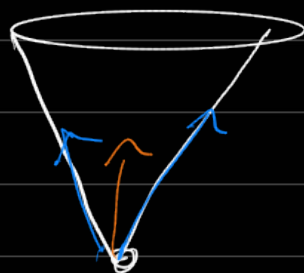
So $T^\mu = \frac{dx^\mu}{dt}$

* This implies λ is t^a variables anywhere λ is not a timelike curve.

* If T^μ vanishes, then $\frac{dx^\mu}{dt} = 0$ or $x^\mu = \text{constant}$ i.e. $\lambda(t) = p$ is not considered a timelike curve.

* Def: λ is said to be a future directed causal curve if $\forall p \in \lambda$, t^a is either future directed or null.

Causal:



$$t = t^a \partial_a : \text{Causal} \Leftrightarrow g(t, t) \leq 0$$

$I^+(p)$: Chronological future of $p \in M$ is defined as the set of events that can be reached by a future directed timelike curve starting from p .



$$I^+(p) = \left\{ q \in M \mid \exists \text{ a future directed timelike curve } \lambda(t) \text{ with } \lambda(0) = p \text{ and } \lambda(1) = q \right\}$$



*

$\mathcal{I}^+(p)$: set of accessible pts.



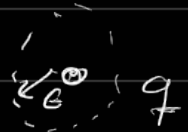
$$\lambda(0) = p$$

Claim: One can always sufficiently perform small deformation of the endpoint of a curve while preserving the timelike nature of λ .

"Proof": Let $\lambda: [0, 1]$ with $\lambda(0) = p, \lambda(1) = q$ and take $B_\epsilon(q)$

for $l \in B_\epsilon(q)$

$$\Rightarrow \|l - q\| < \epsilon$$



Define $\Lambda: [\frac{1}{2}, 1]$ s.t. $\Lambda(0) = \lambda(0) = p$ and $\Lambda(\frac{1}{2}) = \lambda(1) = q$

so that the remaining curve from $[\frac{1}{2}, 1]$ is a curve from q to l .

* Contr: Suppose $\exists$ a curve $\Lambda: [\frac{1}{2}, 1]$ $\Lambda(\frac{1}{2}) = q, \Lambda(1) = l$

Then, $x^\mu(\Lambda(\frac{1}{2}))$

$$\Rightarrow \|x^\mu - y^\mu\| < \epsilon \text{ by def. of } B_\epsilon(q)$$

Now take the vector associated to the curve $v^\mu = \frac{dx^\mu}{dt}$

$\Rightarrow$ From above (Lemma 8.1.1.) we know there exists a smooth nonvanishing timelike vector field t^a . In particular we can locally have a timelike vector for which the curve Λ is tangent to. Hence, a contradiction $\Rightarrow \exists$ a timelike curve which can extend q to l . (Sketchy).

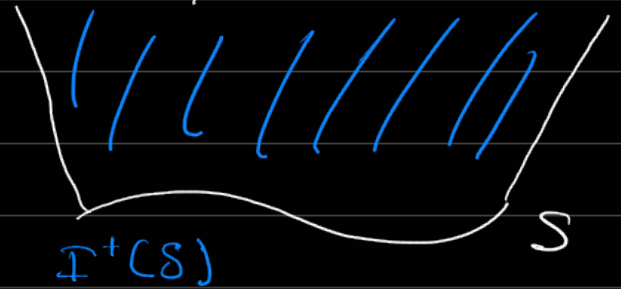
* Follows that:

$$\forall q \in \mathcal{I}^+(p), \exists \mathcal{O}_q \text{ s.t. } \mathcal{O}_q \in \mathcal{I}^+(p)$$

* $\mathcal{I}^+(p)$ is always an open subset of $\mathcal{M}$

→ This is trivial pictorially.

- In general, $p \notin I^+(p)$
- $S \subset M$,
$$I^+(S) = \bigcup_{p \in S} I^+(p)$$



- $I^+(S)$ is also open since the Union of open sets is open

- $I^+[I^+(p)] = I^+(p)$

proof: Let $a \in I^+[I^+(p)]$
then $I^+[I^+(p)] = \bigcup_{q \in I^+(p)} I^+(q)$

$$a \in \bigcup_{q \in I^+(p)} I^+(q) \Rightarrow \exists \tilde{q} \text{ s.t. } a \in I^+(\tilde{q})$$

$$a \in I^+(\tilde{q}) \quad \text{but since } \tilde{q} \in I^+(p)$$

but since $a \in I^+(\tilde{q}) \Rightarrow a \in I^+(p)$
since the curve from p to $\tilde{q}$ to a is timelike.

$$\Rightarrow a \in I^+(p)$$

Let $a \in I^+(p)$, then trivially $a \in I^+[I^+(p)]$
 $I^+[I^+(p)] = I^+(p)$

Def: The Causal future of $p \in M$, denoted $I^+(p)$: $p \in M$: events that can be reached by a future directed causal curve (λ : null or timelike)

$$I^+(p) = \{q \in M \mid \exists \text{ a future directed causal curve } \lambda \text{ with } \lambda(0)=p \text{ and } \lambda(1)=q\}$$

(Closed but not light cone set)

$$\Rightarrow p \in I^+(p)$$



- In flat spacetime $I^+(p)$

is closed.

ex: of non closed set $J^+(p)$

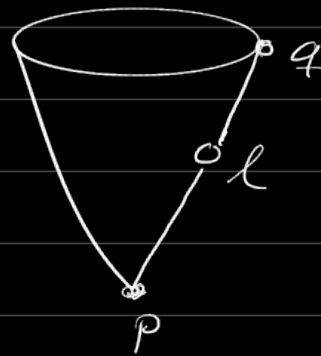
$$M^D \sim \{l\}$$

$\Rightarrow$ No causal curve exists between p and q so

$$q \notin J^+(p)$$

$$q \in J^+(p)$$

$$J^+(s) = \bigcup_{p \in s} J^+(p)$$



Notation:

$$\begin{aligned} p \nmid q &\iff q \in I^+(p) \\ p \nmid q &\iff q \in J^+(p) \end{aligned}$$

$$\partial I^+(p) = \partial J^+(p) \quad (\text{Boundary})$$

$\hookrightarrow$ Null geodesics from p

$$J^+(p) = I^+(p) \cup \partial I^+(p)$$

$\hookrightarrow$ Not valid generally. This is for M^D and doesn't generally hold in general M .

$\hookrightarrow$ Can however be true locally everywhere

i.e.: This is nothing more than a "causal renaming" of the local structure of M . As in field theory, we can always find a local patch to work in: $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$ and add corrections as necessary.

Thm 8.1.2: (M, g_{ab}) be an arbitrary spacetime and $p \in M$. Then $\exists$ a convex nbhd of p $\exists U_a \subset M, p \in U_a$ s.t. $\forall q, r \in U_a \exists \gamma$ which is a geodesic connecting q and r while remaining in U_a . In M^D the geod. condition is replaced with the condition

of a line).

For any such U_α

$$I^+(p)|_U = I^+(p) \cap U_\alpha$$

* Geodesic $\gamma: \mathbb{I} \rightarrow M$

$$T^a = \frac{d\gamma^a}{d\lambda}$$

$$\nabla_T T = 0$$

$$T^a \nabla_a (T^b \partial_b)$$

$$T^a \partial_a T^b \partial_b + T^a T^b \nabla_a (\partial_b)$$

$$= T^a \partial_a T^c \partial_c + T^a T^b \Gamma_{ab}^c \partial_c$$

$$\text{or } T^a \partial_a T^c + T^a T^b \Gamma_{ab}^c = 0$$

$$\frac{dx^a}{d\lambda} \frac{\partial}{\partial x^a} \frac{dx^c}{d\lambda} = \frac{dx^a}{d\lambda} \left(\frac{dx^a}{d\lambda} \right)^{-1} \frac{d^2 x^c}{d\lambda^2}$$

$$\therefore \text{Geodesic eq. } \frac{d^2 x^c}{d\lambda^2} + \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda} \Gamma_{ab}^c = 0$$

$\rightarrow I^+(p)|_U$: consists of all pts reached by a future directed geodesic in U : $I^+(p)|_U = (U, g_{ab})$

$\partial I^+(p)|_U$: generated by future directed null geodesics in U emanating from p .

$$\rightarrow \partial U I^+(p) = \overline{I^+(p)}^U - I^+(p) \quad (\text{Relative to } U)$$

$$\sim U_p, q \in U_p \quad q = \exp_p(X) \quad (p) \quad X \in T_p M$$

locally $q \in I^+(p)|_U$ iff X is a future-directed timelike vector: $g_p(X, X) < 0$

($\Rightarrow$) Let $q \in I^+(p)|_U$, then $\exists \gamma$ s.t.

$$\frac{d\gamma^a}{d\lambda} = X^a \quad \Big| \quad \gamma_x(0) = p, \quad \gamma_x(1) = X^a$$

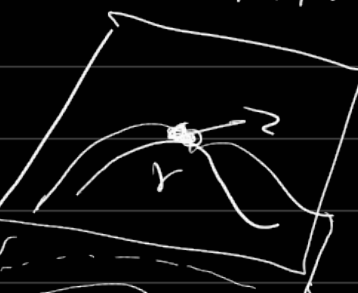
* Geodesic exponential: (M, g) be Riemannian.

$p \in M$, any $X_p \in T_p M$, $\exists!$ $\gamma(t) = \gamma(t; p, X_p)$

s.t.

$$\gamma(0) = p, \quad \dot{\gamma}(0) = X_p$$

$\mathcal{E} \equiv \{ (p, X_p) \mid \gamma(t; p, X_p) \text{ is defined on an interval containing } [0, 1] \}$



$\mathcal{E} = TM$ iff (M, g) is geodesically complete: each geodesic can be defined on $\mathbb{R}$.

→ For any $\lambda \geq 0$:

$$\tilde{\gamma}(t) := \gamma(\lambda t; p, \bar{x}_p)$$

$$\tilde{\gamma}(0) = p, \quad \dot{\tilde{\gamma}}(0) = \lambda \bar{x}_p$$

• If $(p, \bar{x}_p) \in \mathcal{E} \Rightarrow 0 < \lambda < 1, (p, \lambda \bar{x}_p) \in \mathcal{E}$

• If $(p, \bar{x}_p) \notin \mathcal{E}$ one can find $\varepsilon > 0$ s.t.
 $(p, \varepsilon \bar{x}_p) \in \mathcal{E}$.

↳ ex: Suppose $(p, \bar{x}_p) \notin \mathcal{E}$, then

$\exists \alpha < 1$ s.t. the curve ω to $(p, \bar{x}_p)$
 is defined on $\omega: [0, \alpha] \rightarrow M$

• Trivially with $\varepsilon \geq \alpha^{-1}$

$\tilde{\omega} \sim (p, \varepsilon \bar{x}_p) \in \mathcal{E}$ since $\tilde{\omega}: [0, 1] \rightarrow M$
 and any such larger value extends past 1.

def: $\exp: \mathcal{E} \rightarrow M, (p, \bar{x}_p) \mapsto \exp_p(\bar{x}_p)$

$$\exp_p(\bar{x}_p) := \gamma(1; p, \bar{x}_p)$$

$\Rightarrow$ It is the end point of the geodesic section that starts at p in the direction of $\bar{x}_p$ with length $\|\bar{x}_p\|$

expand to $\exp_p(t\bar{x}_p) = \gamma(t; p, \bar{x}_p)$

* Def: we will define the geodesic ball of radius

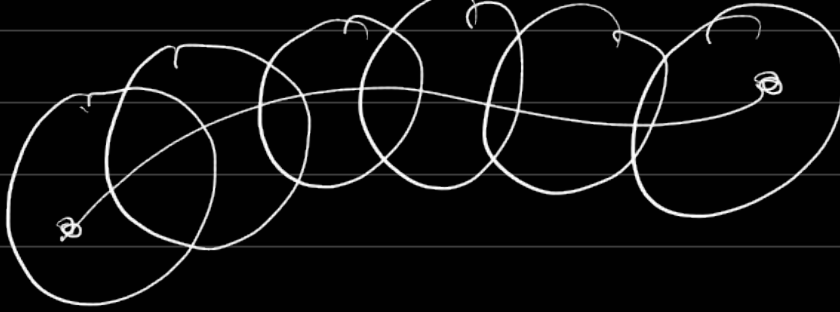
r centered at $p \in M$ as

$$B_p(r) := \exp_p(B_p(0)) \text{ with } B_p(0) \subset T_p M$$

• $\exp_p$ is fiberwise defined.

* $\partial I^+(p)|_U$ is the Boundary of I contained in U so they are indeed null.

• If $q \in \partial I^+(p)$ and λ a causal curve beginning at p and ending at q , we can cover λ by convex normal nbhd's. Since λ is compact we can extract a finite subcover



(Compactness): every open cover has a finite subcover.

Corollary: If $q \in J^+(p) \setminus I^+(p) (M \cup \{ \infty \})$, then any causal curve connecting p and q must be a null geodesic

$$\# S \subset M, \quad J^+(S) \subset \overline{I^+(S)}$$

$$\Rightarrow \overline{I^+(S)} \subset J^+(S) \quad \text{and so} \quad \overline{J^+(S)} = \overline{I^+(S)}$$

$$I^+(S) = \text{int}(J^+(S))$$

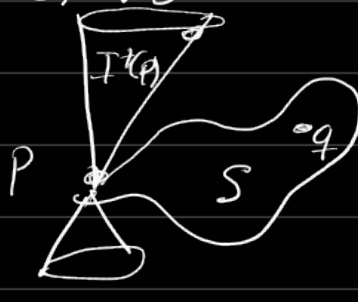
$$\hookrightarrow \partial I^+(S) = \partial J^+(S)$$

No timelike separation within the set

def: $S \subset M$ is said to be achronal if there do not exist $p, q \in S$ s.t. $q \in I^+(p)$ i.e.

$$I^+(S) \cap S = \emptyset$$

ex:



example for $I^+(p)$

example 2: Timelike surfaces/Foliation of S.T.

Thm: Let (M, g) be a time orientable spacetime and let $S \subset M$. Then $\partial I^+(M)$ (if nonempty) is an achronal, 3-dim embedded C^0 -submfd of M .

proof:

